

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

02

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**609**

The correct answer is 609. It's given that the function g is defined by $gx = \frac{x}{a} - 14$, where $a < 0$. Substituting $15a$ for x in function g yields $g15a = \frac{15a}{a} - 14$. This function can be rewritten as $g15a = \frac{15a}{a} - 14$, or $g15a = 15\frac{a}{a} - 14$. Since $a < 0$, it follows that $\frac{a}{a} = -1$. Substituting -1 for $\frac{a}{a}$ in $g15a = 15\frac{a}{a} - 14$ yields $g(15a) = 15(-1) - 14$, or $g(15a) = -29$. Similarly, substituting $7a$ for x in function g yields $g7a = \frac{7a}{a} - 14$. This function can be rewritten as $g7a = \frac{7a}{a} - 14$, or $g7a = 7\frac{a}{a} - 14$. Since $a < 0$, it again follows that $\frac{a}{a} = -1$. Substituting -1 for $\frac{a}{a}$ in $g7a = 7\frac{a}{a} - 14$ yields $g(7a) = 7(-1) - 14$, or $g(7a) = -21$. Therefore, $g(15a) = -29$ and $g(7a) = -21$. Thus, the product of $g(15a)$ and $g(7a)$ is $(-29)(-21)$, or 609.

Q2**A****A. 16**

Choice A is correct. The equation $fx = 1.84^{\frac{x}{4}}$ can be rewritten as $fx = 1.84^{\frac{1}{4}x}$, which is equivalent to $fx = 1.84^{\frac{1}{4}x}$, or approximately $fx = 1.16467^x$. Since it's given that $fx = 1.84^{\frac{x}{4}}$ can be rewritten as $fx = 1 + \frac{p}{100}^x$, where p is a constant, it follows that $1 + \frac{p}{100}$ is approximately equal to 1.16467. Therefore, $\frac{p}{100}$ is approximately equal to 0.16467. It follows that the value of p is approximately equal to 16.467. Of the given choices, 16 is closest to the value of p .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

C

C. $fx = -6x - 5^2 + 24$

Choice C is correct. The given equation is quadratic. The maximum value of a function defined by a quadratic equation can be displayed as a constant in the vertex form of the equation, $f(x) = a(x - h)^2 + k$, where the maximum value of the function is k , which occurs when $x = h$, and a is a constant. The given equation can be rewritten in this form by completing the square. To complete the square, the given equation can be rewritten as $f(x) = -6(x^2 - 10x) - 126$, which is equivalent to $f(x) = -6(x^2 - 10x + 25) - 126 + 6(25)$, or $f(x) = -6(x - 5)^2 + 24$. This equation is in vertex form, where $a = -6$, $h = 5$, and $k = 24$. Therefore, $f(x) = -6(x - 5)^2 + 24$ displays the maximum value, 24, of the function as a constant.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This form displays one of the zeros, not the maximum value, of the function as a constant.

Choice D is incorrect. This form displays the zeros, not the maximum value, of the function as constants.

Q4

C

C. $qx = 140.55^x$

Choice C is correct. Since the value of qx decreases by a fixed percentage, 45%, for every increase in the value of x by 1, the function q is a decreasing exponential function. A decreasing exponential function can be written in the form $qx = a1 - \frac{p}{100}^x$, where a is the value of q_0 and the value of qx decreases by $p\%$ for every increase in the value of x by 1. If $q_0 = 14$, then $a = 14$. Since the value of qx decreases by 45% for every increase in the value of x by 1, $p = 45$. Substituting 14 for a and 45 for p in the equation $qx = a1 - \frac{p}{100}^x$ yields $qx = 141 - \frac{45}{100}^x$, which is equivalent to $qx = 141 - 0.45^x$, or $qx = 140.55^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. For this function, the value of qx increases, rather than decreases, by 45% for every increase in the value of x by 1.

Q5**D**

D. $-\frac{5}{2}$

Choice D is correct. It's given that $f(x) = x^2 - 14x + 19$, which can be rewritten as $f(x) = x^2 + 5x - 266$. Since the coefficient of the x^2 -term is positive, the graph of $y = f(x)$ in the xy -plane opens upward and reaches its minimum value at its vertex. The x -coordinate of the vertex is the value of x such that $f(x)$ reaches its minimum. For an equation in the form $f(x) = ax^2 + bx + c$, where a , b , and c are constants, the x -coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 + 5x - 266$, $a = 1$, $b = 5$, and $c = -266$. It follows that the x -coordinate of the vertex is $-\frac{5}{2}$, or $-\frac{5}{2}$. Therefore, $f(x)$ reaches its minimum when the value of x is $-\frac{5}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x -value of the midpoint between the two x -intercepts of the parabola. Since it's given that $f(x) = x^2 - 14x + 19$, it follows that the two x -intercepts of the graph of $y = f(x)$ in the xy -plane occur when $x = 14$ and $x = -19$, or at the points $(14, 0)$ and $(-19, 0)$. The midpoint between two points, (x_1, y_1) and (x_2, y_2) , is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Therefore, the midpoint between $(14, 0)$ and $(-19, 0)$ is $\left(\frac{14 + (-19)}{2}, \frac{0 + 0}{2}\right)$, or $\left(-\frac{5}{2}, 0\right)$. It follows that $f(x)$ reaches its minimum when the value of x is $-\frac{5}{2}$.

Choice A is incorrect. This is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane.

Choice B is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6

D

D. 11.00

Choice D is correct. The value of $p(12)$ depends on the value of the constant c , so the value of c must first be determined. It is given that $p(c) = 10$. Based on the definition of p , it follows that:

$$p(c) = \frac{(c - c)^2 + 160}{2c} = 10$$

$$\frac{160}{2c} = 10$$

$$2c = 16$$

$$c = 8$$

This means that $p(x) = \frac{(x - 8)^2 + 160}{16}$ for all values of x . Therefore:

$$p(12) = \frac{(12 - 8)^2 + 160}{16}$$

$$= \frac{16 + 160}{16}$$

$$= 11$$

Choice A is incorrect. It is the value of $p(8)$, not $p(12)$. Choices B and C are incorrect. If one of these values were correct, then $x = 12$ and the selected value of $p(12)$ could be substituted into the equation to solve for c . However, the values of c that result from choices B and C each result in $p(c) < 10$.

Q7**D**

- D. The estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

Choice D is correct. The y -intercept of a graph in the ty -plane is the point where $t = 0$. For the given function f , the y -intercept of the graph of $y = ft$ in the ty -plane can be found by substituting 0 for t in the equation $y = 8,0000.65^t$, which gives $y = 8,0000.65^0$. This is equivalent to $y = 8,0001$, or $y = 8,000$. Therefore, the y -intercept of the graph of $y = ft$ is 0, 8,000. It's given that the function f models the number of coupons a company sent to their customers at the end of each year. Therefore, ft represents the estimated number of coupons the company sent to their customers at the end of each year. It's also given that t represents the number of years since the end of 1998. Therefore, $t = 0$ represents 0 years since the end of 1998, or the end of 1998. Thus, the best interpretation of the y -intercept of the graph of $y = ft$ is that the estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q8**D**

- D. $rm = 90,0001.06^{\frac{m}{12}}$

Choice D is correct. It's given that the function p models the population of Lowell t years after a census. Since there are 12 months in a year, m months after the census is equivalent to $\frac{m}{12}$ years after the census. Substituting $\frac{m}{12}$ for t in the equation $pt = 90,0001.06^t$ yields $p^{\frac{m}{12}} = 90,0001.06^{\frac{m}{12}}$. Therefore, the function r that best models the population of Lowell m months after the census is $rm = 90,0001.06^{\frac{m}{12}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q9**D****D. Day 12**

Choice D is correct. The number of bacteria per milliliter is doubling each day. For example, from day 1 to day 2, the number of bacteria increased from 2.5×10^5 to 5.0×10^5 . At the end of day 3 there are 10^6 bacteria per milliliter. At the end of day 4, there will be $10^6 \times 2$ bacteria per milliliter. At the end of day 5, there will be $(10^6 \times 2) \times 2$, or $10^6 \times (2^2)$ bacteria per milliliter, and so on. At the end of day d , the number of bacteria will be $10^6 \times (2^{d-3})$. If the number of bacteria per milliliter will reach 5.12×10^8 at the end of day d , then the equation $10^6 \times (2^{d-3}) = 5.12 \times 10^8$ must hold. Since 5.12×10^8 can be rewritten as 512×10^6 , the equation is equivalent to $2^{d-3} = 512$. Rewriting 512 as 2^9 gives $d - 3 = 9$, so $d = 12$. The number of bacteria per milliliter would reach 5.12×10^8 at the end of day 12.

Choices A, B, and C are incorrect. Given the growth rate of the bacteria, the number of bacteria will not reach 5.12×10^8 per milliliter by the end of any of these days.

Q10**D****D. Neither I nor II**

Choice D is correct. A function of the form $f(x) = ab^x + c$, where $a < 0$ and $b > 1$, is a decreasing function. Both of the given functions are of this form; therefore, both are decreasing functions. If a function f is decreasing as the value of x increases, the corresponding value of $f(x)$ decreases; therefore, the function doesn't have a minimum value. Thus, neither of the given functions has a minimum value.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

